CSE320: Data Communications

Assignment 3 [Summer 2025] (Version 1.0)

Time Division Multiplexing (TDM)

1. A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?
2. Five channels are multiplexed using TDM. Each channel can process 1000 bits per second. If 5 bits at a time are to be multiplexed using the concept of TDM with 2 synchronizing bits, then find the following.
   (a) Input rate & input duration
   (b) Frame size, frame rate & frame duration
   (c) Output rate & output duration
   (d) Output slot duration
3. Five channels are multiplexed using TDM. Each channel can process 1000 pages per 5 seconds. Each page has 20 lines with 40 characters. If 5 lines at a time are to be multiplexed using the concept of TDM with 5 synchronizing bits, then find the following.
   (a) Input rate & input duration
   (b) Frame size, frame rate & frame duration
   (c) Output rate & output duration
4. Two channels, one with a bit rate of 100 kbps and another with a bit rate of 200 kbps, are to be multiplexed. How can this be achieved? Show the scheme. What is the frame rate? What is the frame duration? What is the bit rate of the link?
5. Six channels, three with a bit rate of 240 kbps and three with a bit rate of 120 kbps, are to be multiplexed. If 5 characters from each channel are to be multiplexed with three synchronization bits, then answer the following questions.
   (a) Show how this can be achieved.
   (b) What is the input rate and input duration?
   (c) What will be the frame size, frame rate & frame duration?
   (d) What will be the output rate & output duration?
   (e) What will be the output slot duration?
6. Five channels, two with a bit rate of 240 kbps and three with a bit rate of 180 kbps, are to be multiplexed, with one synchronization bit and two interleaved bits. Answer the following questions.
   (a) Show how this can be achieved.
   (b) What is the input rate and input duration?
   (c) What will be the frame size, frame rate & frame duration?
   (d) What will be the output rate & output duration?
   (e) What will be the output slot duration?
7. A control system has 8 devices. In the first time slot, only devices 1, 3, 5, 6, and 8 have data. In the second time slot, devices 1, 2, 4, 5, and 7 have data.
   (a) What TMD technique should be used here (synchronous or statistical), and why?
   (b) Sketch a figure of the TDM process and show the frames.
   (c) Sketch a figure of the other TDM process and show the frames.
   (d) When should the other TDM process be used?

Error Detection & Correction

8. Consider the following coding scheme for error detection and correction.

Dataword	Codeword
00	00001111
01	01010011
10	10110100
11	11111001

   - Write the scheme in *C(n, k)* format. How many redundant bits are used in this scheme?
   - How many datawords and codewords are possible in this scheme? How many of the codewords are valid, and how many are invalid (extra)?
   - Find the maximum error detecting and correcting capability of this scheme.
   - Suppose the receiver receives two bit streams: 00011101 & 10111100. Find out whether they are correct or corrupted. If corrupted, find out whether it is possible to recover the original dataword.

9. Let **X** be the last 4 digits of your ID, and **D** and **G** be the binary representations of X % 256, and X % 16, respectively. Consider **D** as the dataword and **G** as the generator.

   - Using **binary division**, find the codeword that is to be transmitted using CRC scheme.
   - Suppose an error occurs during transmission, and the **5th** bit form **MSB** is flipped. Can the CRC scheme detect this error? Show calculations.

10. Suppose you are the recipient of a data transmission system which uses CRC scheme for error detection and correction. In the following table we have four codewords received by you. The divisor/generator for each codeword is mentioned on the right column.

Codeword	Generator
100110100	1001
100110010	1001
100100110	1100
100101110	1110

    - Write the scheme in *C(n, k)* format. How many redundant bits are used in this scheme?
    - How many datawords and codewords are possible in this scheme? How many of the codewords are valid, and how many are invalid (extra)?
    - If your student ID is **even**, find out whether the **first** two codewords are correct or corrupted using **Polynomial division**.
    - If your student ID is **odd**, find out whether the **last** two codewords are correct or corrupted using **Polynomial division**.